Maternal vaginal microflora during pregnancy and the risk of asthma hospitalization and use of antiasthma medication in early childhood.

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BACKGROUND: Infants with wheezing and allergic diseases have a microflora that differs from that of healthy infants. The fetus acquires microorganisms during birth when exposed to the maternal vaginal microflora. It is therefore conceivable that the maternal vaginal microflora might influence the establishment of the infant flora and, as a consequence, the development of wheezing and allergic diseases. OBJECTIVE: We sought to study the associations between the composition of the maternal vaginal microflora and the development of wheezing and asthma in childhood. METHODS: We performed a population-based cohort study in Denmark. Vaginal samples for bacterial analysis were obtained during pregnancy. A total of 2927 women (80% of the invited women) completed the study and had 3003 live infants. Infant wheezing was assessed as one or more hospitalizations for asthma between 0 and 3 years of age. Asthma was assessed as use of 3 or more packages of antiasthma medication between 4 and 5 years of age. RESULTS: Maternal vaginal colonization with Ureaplasma urealyticum during pregnancy was associated with infant wheezing (odds ratio [OR], 2.0; 95% CI, 1.2-3.6), but not with asthma, during the fifth year of life. Maternal colonization with staphylococci (OR, 2.2; 95% CI, 1.4-3.4) and use of antibiotics in pregnancy (OR, 1.7; 95% CI, 1.1-2.6) were associated with asthma during the fifth year of life. CONCLUSION: The composition of the maternal vaginal micro-flora might be associated with wheezing and asthma in the offspring up to 5 years of age.